

系统开发工具实验 1

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1 实验内容

- 1、课程概览与版本控制 (Git);
- 2、Latex 文档编辑;

2 实验过程

2.1 克隆本课程网站的仓库

`git clone`

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea
$ git clone https://github.com/missing-semester-cn/missing-semester-cn.github.io
.git
Cloning into 'missing-semester-cn.github.io'...
remote: Enumerating objects: 3194, done.
remote: Counting objects: 100% (3194/3194), done.
remote: Compressing objects: 100% (1126/1126), done.
remote: Total 3194 (delta 2040), reused 2735 (delta 2033), pack-reused 0 (from 0)
Receiving objects: 100% (3194/3194), 15.44 MiB | 5.75 MiB/s, done.
Resolving deltas: 100% (2040/2040), done.
```

2.2 将版本历史可视化并进行探索

`git log --all --graph --decorate`

```
MINGW64/c/Users/Lenovo/Desktop/missing-semester-cn.github.io

Lenovo@LAPTOP-B037UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git log --all --graph --decorate
* commit af054fa1aea2f2599e4474d96b63f73dd9bd145f (HEAD -> master, origin/master, origin/HEAD)
Merge: dd3f3dd 9baa48c
Author: Lingfeng_Ai <hanxiaomax@gmail.com>
Date: Fri Aug 16 06:54:16 2024 +0800

    Merge pull request #172 from pspdada/master

    Thank you so much

* commit 9baa48c778012164179e4e60725418941f41743b
Author: psp_dada <i824427006@qq.com>
Date: Thu Aug 15 02:07:36 2024 +0800

    remove irrelevant text

* commit f5df7de89dc7712483665cc6fe8a787aafbef9bf
Author: psp_dada <i824427006@qq.com>
Date: Thu Aug 15 01:46:12 2024 +0800

    fix wrong index

* commit ef9a2f75409ff7746c03f6233066e3d2c634cd12
...skipping...
* commit af054fa1aea2f2599e4474d96b63f73dd9bd145f (HEAD -> master, origin/master, origin/HEAD)
Merge: dd3f3dd 9baa48c
Author: Lingfeng_Ai <hanxiaomax@gmail.com>
Date: Fri Aug 16 06:54:16 2024 +0800

    Merge pull request #172 from pspdada/master

    Thank you so much

* commit 9baa48c778012164179e4e60725418941f41743b
Author: psp_dada <i824427006@qq.com>
Date: Thu Aug 15 02:07:36 2024 +0800

    remove irrelevant text

* commit f5df7de89dc7712483665cc6fe8a787aafbef9bf
Author: psp_dada <i824427006@qq.com>
Date: Thu Aug 15 01:46:12 2024 +0800

    fix wrong index
```

2.3 是谁最后修改了 README.md 文件 ? (提示: 使用 git log 命令并添加合适的参数)

git log -1 README.md

-x 选项: 查看最新的 x 次提交或特定文件的版本信息

```
MINGW64/c/Users/Lenovo/Desktop/missing-semester-cn.github.io

Lenovo@LAPTOP-B037UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git log -1 README.md
commit d898852ef0604cf918bab7f39c63a53932c845d8
Author: yuzq <yuzq@sunwayworld.com>
Date: Thu Jun 6 14:43:07 2024 +0800

    将readme文件中的url的绝对路径改为相对路径, 不用重复访问github, 利于分享传播
```

2.4 最后一次修改 __config.yml 文件中 collections: 行时的提交信息是什么 ? (提示: 使用 git blame 和 git show)

git blame __config.yml | grep collections

git show a88b4eac

```

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git blame _config.yml | grep collections
a88b4eac (Anish Athalye 2020-01-17 15:26:30 -0500 18) collections:

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git show a88b4eac
commit a88b4eac326483e29bdac5ee0a39b180948ae7fc
Author: Anish Athalye <me@anishathalye.com>
Date: Fri Jan 17 15:26:30 2020 -0500

    Redo lectures as a collection

diff --git a/2020/index.html b/2020/index.html
deleted file mode 100644
index 153ddc8..0000000
--- a/2020/index.html
+++ /dev/null
@@ -1,5 +0,0 @@
-----
-!layout: redirect
-redirect: /
-title: Missing Semester 2020
-----
diff --git a/2019/automation.md b/_2019/automation.md
similarity index 100%
rename from 2019/automation.md
rename to _2019/automation.md
diff --git a/2019/backups.md b/_2019/backups.md
similarity index 100%
...skipping...
commit a88b4eac326483e29bdac5ee0a39b180948ae7fc
Author: Anish Athalye <me@anishathalye.com>
Date: Fri Jan 17 15:26:30 2020 -0500

    Redo lectures as a collection

diff --git a/2020/index.html b/2020/index.html
deleted file mode 100644
index 153ddc8..0000000
--- a/2020/index.html
+++ /dev/null
@@ -1,5 +0,0 @@
-----
-!layout: redirect
-redirect: /
-title: Missing Semester 2020
-----
diff --git a/2019/automation.md b/_2019/automation.md
similarity index 100%
rename from 2019/automation.md
rename to _2019/automation.md
diff --git a/2019/backups.md b/_2019/backups.md
similarity index 100%

```

2.5 从 GitHub 上克隆某个仓库，修改一些文件。并查看当前状态

修改文件 index.md

先输入命令查看当前状态: git status

```

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.md

no changes added to commit (use "git add" and/or "git commit -a")

```

2.6 当您使用 git stash 会发生什么？

输入命令：git stash

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git stash
Saved working directory and index state WIP on master: 6605024 add password123 to file
```

通过结果可以看出：当你使用 git stash 时，Git 会将你当前工作目录和暂存区的修改保存到一个堆栈中，并将你的工作区恢复到最后一次提交的状态。

2.7 当您执行 git log -all -oneline 时会显示什么？

输入命令：git log -all -oneline

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git log --all --oneline
2c5cf43 (refs/stash) WIP on master: 6605024 add password123 to file
5481c2a index on master: 6605024 add password123 to file
6605024 (HEAD -> master) add password123 to file
af054fa (Origin/master, origin/HEAD) Merge pull request #172 from pspdada/master
9baa48c remove irrelevant text
f5df7de fix wrong index
ef9a2f7 fix typo
dd3f3dd Merge pull request #171 from HowieChih/for-better-understanding
8a26b4a 更新#课程概览与 shell##一个功能全面又强大的工具关于修改亮度文件报错的翻译
4e2ff43 更新#课程概览与 shell##一个功能全面又强大的工具关于修改亮度文件报错的翻译
d284d3e Merge pull request #170 from crosscap/typo-fix
6e27b4f fix some typo in version-control.md
f3773ac Merge pull request #169 from xjzh123/patch-1
ede827e Merge pull request #168 from DXShelley/master
45897e1 修复multiple错译为“乘积”
6221bbc Merge branch 'master' of https://pi-gitserver.kooldns.cn/dxshelley/missing-semester-cn 合并
de98852 将readme文件中的url的绝对路径改为相对路径，不用重复访问github，利于分享传播
3417047 'test'
d78268f Merge pull request #162 from moralok/master
:...skipping...
2c5cf43 (refs/stash) WIP on master: 6605024 add password123 to file
5481c2a index on master: 6605024 add password123 to file
6605024 (HEAD -> master) add password123 to file
```

通过结果可以看出：执行 git log -all -oneline 会显示所有分支上的提交记录，以简洁的一行格式列出提交信息。

2.8 通过 git stash pop 命令来撤销 git stash 操作，什么时候会用到这一技巧？

输入命令来撤销 git stash 操作: git stash pop

```

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git stash pop
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (2c5cf43f38a4883c5912707856313118b9321900)

```

通过结果可以看出：使用 `git stash pop` 时，Git 会从堆栈中恢复之前保存的修改，并尝试将这些修改应用到当前的工作目录。

如果应用过程中出现冲突，你需要手动解决。这一技巧通常用于当你需要切换到其他分支或执行其他操作时暂时保存当前的工作状态。

2.9 与其他的命令行工具一样, Git 也提供了一个名为 `/.gitconfig` 配置文件 (或 dotfile)。请在 `/.gitconfig` 中创建一个别名

进入文件：`vim /.gitconfig` 在文件中创建别名：

```
graph = log -all -graph -decorate -oneline
```



```

MINGW64/c:/Users/Lenovo/Desktop/missing-semester-cn.github.io
[credential "https://gitee.com"]
  provider = generic
[user]
  email = 1013336369@qq.com
  name = 廉嘉蕊
[core]
  excludesfile = C:/Users/Lenovo/.gitignore
[alias]
  graph = log --all --graph --decorate --oneline

```

2.10 您在运行 `git graph` 时，您可以得到 `git log -all -graph -decorate -oneline` 的输出结果

```
git graph
```

```

lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ vim ~/.gitconfig
lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git graph
* 6605024 (HEAD -> master) add password123 to file
* af054fa (origin/master, origin/HEAD) Merge pull request #172 from pspada/master
* 9baa48c remove irrelevant text
* f5df7de fix wrong index
* ef9a2f7 fix typo
* dd3f3dd Merge pull request #171 from HowieChih/for-better-understanding
* 8e26b4a 更新#课程概览与 shell## 一个功能全面又强大的工具关于修改亮度文件报错的翻译
* 4e2ff43 更新#课程概览与 shell## 一个功能全面又强大的工具关于修改亮度文件报错的翻译
* d284d3e Merge pull request #170 from crosscap/typo-fix
* 6e27b4f fix some typo in version-control.md
* f3773ac Merge pull request #169 from xjzh123/patch-1
* 45897e1 修复multiple错译为“乘积”
* ede827e Merge pull request #168 from DXShelley/master
...skipping...
* 6605024 (HEAD -> master) add password123 to file
* af054fa (origin/master, origin/HEAD) Merge pull request #172 from pspada/master
* 9baa48c remove irrelevant text
* f5df7de fix wrong index
* ef9a2f7 fix typo
* dd3f3dd Merge pull request #171 from HowieChih/for-better-understanding
* 8e26b4a 更新#课程概览与 shell## 一个功能全面又强大的工具关于修改亮度文件报错的翻译
* 4e2ff43 更新#课程概览与 shell## 一个功能全面又强大的工具关于修改亮度文件报错的翻译
* d284d3e Merge pull request #170 from crosscap/typo-fix
* 6e27b4f fix some typo in version-control.md
* f3773ac Merge pull request #169 from xjzh123/patch-1

```

2.11 您可以通过执行 `git config --global core.excludesfile ~/.gitignore_global` 在 `~/.gitignore_global` 中创建全局忽略规则。配置您的全局 `gitignore` 文件来自动忽略系统或编辑器的临时文件，例如 `.DS_Store`；

`git config --global core.excludesfile ~/.gitignore .DS_Store`

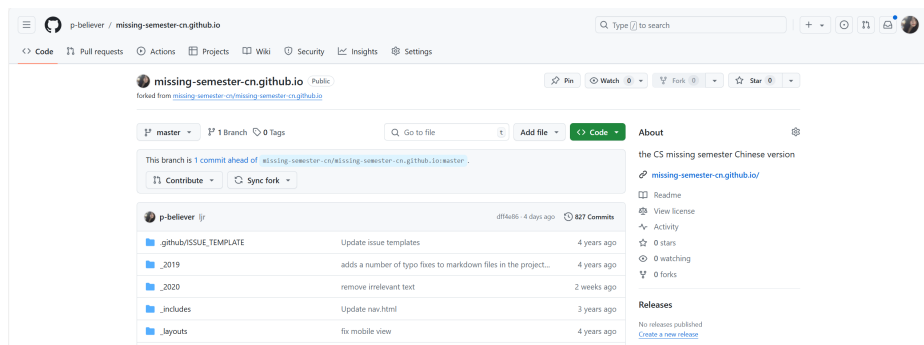
```

lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/missing-semester-cn.github.io (master)
$ git config --global core.excludesfile ~/.gitignore .DS_Store

```

2.12 Fork 本课程网站的仓库，找找有没有错别字或其他可以改进的地方

fork 本课程网站的仓库



2.13 更新仓库并推上去

克隆 fork 后的仓库，并修改其中的一处，再将其推到 github

```
MINGW64/c:/Users/Lenovo/Desktop/大三上/missing-semester-cn.github.io

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git add .

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git commit -m "ljr"
[master dff4e86] ljr
1 file changed, 7 insertions(+), 7 deletions(-)

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git push
fatal: unable to access 'https://github.com/p-believer/missing-semester-cn.github.io.git/': Failure when receiving data from the peer

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git push
fatal: unable to access 'https://github.com/p-believer/missing-semester-cn.github.io.git/': Failed to connect to github.com port 443 after 21040 ms: couldn't connect to server

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git push -u origin master
fatal: unable to access 'https://github.com/p-believer/missing-semester-cn.github.io.git/': Failed to connect to github.com port 443 after 21127 ms: couldn't connect to server

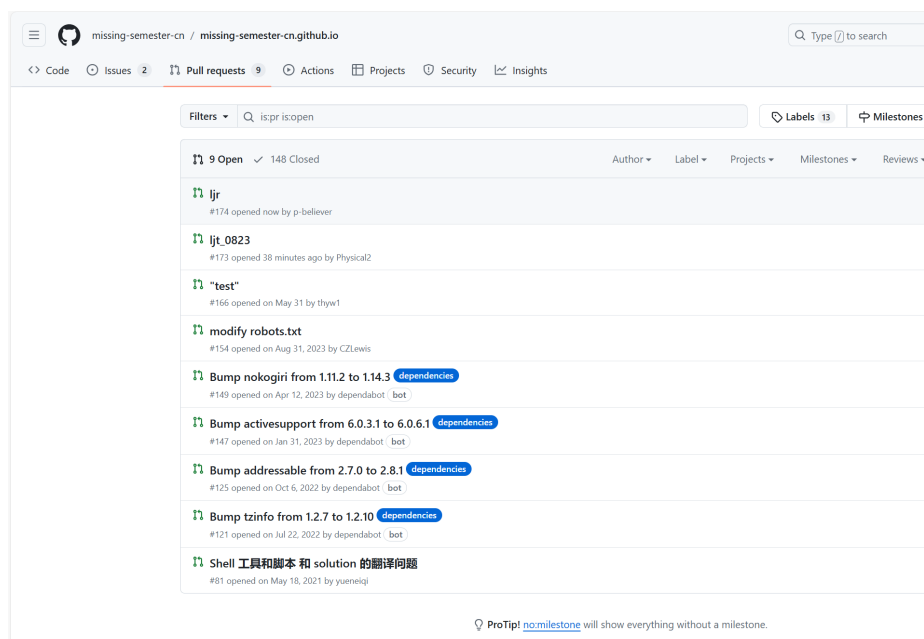
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git push -u origin master
fatal: unable to access 'https://github.com/p-believer/missing-semester-cn.github.io.git/': Failed to connect to github.com port 443 after 21149 ms: couldn't connect to server

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 323 bytes | 161.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/p-believer/missing-semester-cn.github.io.git
  af054fa..dff4e86  master -> master
branch 'master' set up to track 'origin/master'.

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/大三上/missing-semester-cn.github.io (master)
$ |
```

2.14 在 GitHub 上发起拉取请求 (Pull Request)

在 GitHub 上发起拉取请求，之后可以在本课程网站的仓库中看到拉取请求



2.15 初始化仓库

git init

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (master)
$ git init
Reinitialized existing Git repository in C:/Users/Lenovo/Desktop/.idea/missing-semester-cn.github.io/.git/
```

2.16 查看本地分支

git branch

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (ma
ster)
$ git branch
* master
```

2.17 创建新的分支

`git branch new`

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (ma
ster)
$ git branch new
```

2.18 重命名分支并进入该分支

`git branch -m renew`

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (ma
ster)
$ git branch -m renew

Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (re
new)
$
```

2.19 清理未跟踪的文件和目录

`git clean -fd`

```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (re
new|REBASE 29/662)
$ git clean -fd
```

2.20 交互式暂存文件

`git add -p`


```
Lenovo@LAPTOP-B0J7UK16 MINGW64 ~/Desktop/.idea/missing-semester-cn.github.io (re
new|REBASE 29/662)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
credential.https://gitee.com.provider=generic
user.email=1013336369@qq.com
user.name=康嘉蕊
core.excludesfile=C:/Users/Lenovo/.gitignore
alias.graph=log --all --graph --decorate --oneline
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
:...skipping...
```

3 实验总结

进行 Git 命令实验是深入理解 Git 工作原理和功能的一个好方法。通过本次实验，我掌握了 git 相关的基本知识，深入理解 Git 的工作机制和管理版本控制的能力，并学会了使用 latex，受益匪浅。